Model Evaluation Framework

Complete guide and tools for evaluating LLMs and AI models.

📚 Documentation

MODEL-EVALUATION-GUIDE.md - Comprehensive guide covering:

- Automated metrics (BLEU, ROUGE, BERTScore, Perplexity)
- Human evaluation methods
- Benchmark datasets (GLUE, SQuAD, HumanEval, GSM8K)
- Tools & frameworks (LangChain, RAGAS, TruLens)
- Production monitoring
- Best practices

DEPLOYMENT-READINESS.md - When to deploy a model

- Performance criteria (benchmarks, baseline comparison)
- Business requirements (ROI, UX thresholds)
- Safety checks (error rates, bias, adversarial robustness)
- Operational readiness (monitoring, rollback, A/B testing)
- Complete deployment checklist with code examples

LLM-USAGE-GOVERNANCE.md - **How to limit LLM usage across teams** 🆕

- API Gateway with rate limiting and quotas
- Token budget system with alerts
- Cost tracking and attribution
- Priority queuing for critical teams
- Usage policies and approval workflows
- Monitoring dashboard and admin interface

🚀 Quick Start

Setup

```
cd model-evaluation
pip install -r requirements.txt

# Set API keys
export GROQ_API_KEY="your_groq_key"
export ANTHROPIC_API_KEY="your_anthropic_key"
```

Run Evaluation

```
python creative-hub-eval.py
```

📊 Example Output

```
Starting Creative Automation Hub Model Evaluation
Test cases: 3
Models: Groq (llama-3.3-70b) vs Anthropic (claude-sonnet-4)

[1/3] Testing: Write a tweet about sustainable fashion...
[2/3] Testing: Write a blog intro about remote work...
[3/3] Testing: Create ad copy for a project management...

============================================================
EVALUATION SUMMARY
============================================================

Groq (llama-3.3-70b):
  ROUGE-1:      0.654 ± 0.082
  ROUGE-L:      0.588 ± 0.091
  BERTScore F1: 0.892 ± 0.034
  Avg Time:     2.14s

Anthropic (claude-sonnet-4):
  ROUGE-1:      0.698 ± 0.075
  ROUGE-L:      0.621 ± 0.088
  BERTScore F1: 0.915 ± 0.028
  Avg Time:     3.87s

============================================================
Winner: Anthropic
============================================================
```

🧪 Evaluation Metrics

Automated Metrics

Metric	Purpose	Range	Higher = Better
ROUGE-1	Word overlap	0-1	✅
ROUGE-L	Longest common subseq	0-1	✅
BERTScore	Semantic similarity	0-1	✅
Perplexity	Language model quality	0-∞	❌ (lower better)

LLM-as-Judge Criteria

- **Engagement**: Is it engaging and shareable?
- **Clarity**: Is the message clear?
- **Professionalism**: Appropriate tone?

- **Call-to-Action**: Does it inspire action?

📝 Custom Test Cases

Add your own test cases:

```python
from creative_hub_eval import TestCase

test_case = TestCase(
    prompt="Your prompt here",
    content_type="social",  # or "blog", "ad"
    tone="professional",    # or "casual", "friendly"
    reference="Expected output here",
    criteria={
        "criterion_1": "Description",
        "criterion_2": "Description"
    }
)
```

🔍 Evaluation Methods

1. Automated Metrics

- Fast and cheap
- Consistent
- Good for regression testing

2. LLM-as-Judge

- Scalable
- Evaluates nuanced criteria
- Decent correlation with human judgment

3. Human Evaluation

- Gold standard
- Slow and expensive
- Use for final validation

📈 Tracking Over Time

```python
# Run weekly evaluations
results_week_1 = evaluator.run_batch(test_cases)
results_week_2 = evaluator.run_batch(test_cases)

# Compare
improvement = (
    results_week_2['bert_f1_mean'] -
```

```
    results_week_1['bert_f1_mean']
)
```

🛠️ Tools Used

- **bert-score**: Semantic similarity via BERT embeddings
- **rouge-score**: N-gram overlap for summarization
- **Groq API**: Fast LLM inference
- **Anthropic API**: High-quality generation

📚 Additional Resources

- GLUE Benchmark
- HuggingFace Evaluate
- LangChain Evaluators
- RAGAS Documentation

🎯 Use Cases

Creative Automation Hub

- Evaluate text generation quality
- Compare Groq vs Anthropic
- A/B test different prompts

General LLM Projects

- Model selection
- Prompt engineering
- Fine-tuning validation
- Production monitoring

🤝 Contributing

Add more test cases, metrics, or evaluation methods:

1. Add test cases to `create_test_suite()`
2. Implement new metrics in `evaluate_single()`
3. Add criteria to `llm_as_judge()`

📖 Learn More

See MODEL-EVALUATION-GUIDE.md for:

- Detailed metric explanations
- Benchmark datasets
- Production monitoring setup
- Statistical significance testing
- Cost analysis

💡 Tips

1. **Start simple**: Use automated metrics first
2. **Add LLM-as-judge**: For nuanced criteria
3. **Validate with humans**: Sample 10-20% for human eval
4. **Track over time**: Monitor for regressions
5. **Consider cost**: Balance quality vs API costs

📊 Example Results

Groq Pros:

- ✅ 2x faster (2.1s vs 3.9s)
- ✅ Cheaper ($0.05 vs $0.15 per 1K tokens)

Anthropic Pros:

- ✅ Higher quality (0.915 vs 0.892 BERTScore)
- ✅ Better ROUGE scores

Choose based on:

- Groq: Speed and cost matter
- Anthropic: Quality matters most